

EXPOSÉ

Portfolio Adaptation Across the Wealth Distribution: Evidence from U.S. DFA Local Projections

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1 Research Question

Wealth groups differ substantially in their exposure to financial markets, their willingness to take risk, and their ability to adapt portfolios when asset prices move, with wealthier households typically holding more risky assets and facing lower constraints. While much is known about how individual investors behave in micro-level data, far less is understood about how groups of households across the wealth distribution adapt their portfolios in real time when equity values change, particularly in a setting where balance-sheet components can be tracked consistently at a medium frequency. This gap is important because differences in rebalancing behavior shape how valuation shocks propagate through balance sheets and influence long-run, absolute wealth accumulation despite ongoing wealth inequality. The research question is therefore whether wealthier groups rebalance their equity positions more promptly when confronted with unforeseen valuation movements. Using the U.S. Distributional Financial Accounts (DFA), which provides quarterly balance-sheet data for distinct wealth groups, the analysis examines how they respond to group-specific valuation shocks particularly in corporate equities and mutual fund holdings, relying on local projections to capture flexible, horizon-specific adjustments.

2 Relevant Literature

Research on household portfolios consistently shows that wealth groups differ sharply in both their asset composition and their exposure to valuation shocks. Using long-run SCF microdata,

Bricker et al. (2019) document that most U.S. households hold portfolios dominated by housing and financial assets, while only the upper tail, particularly the top 1 percent, holds substantial business equity and diversified financial market assets. As they emphasize, “most families hold a relatively small share of assets in financial markets and instead hold primarily housing assets,” whereas the top of the distribution is characterized by large positions in private businesses and equities. These structural differences imply that unexpected changes in asset values propagate very differently across the wealth distribution, shaping both the magnitude of valuation shocks households experience and the channels through which they can adapt.

Evidence from administrative data further shows that households actively respond to these shocks, though the intensity and direction of rebalancing vary systematically with wealth and sophistication. Calvet et al. (2008), using comprehensive Swedish registry data, find that households partially offset idiosyncratic fluctuations in their risky-asset share, with active rebalancing “on average offset[ting] about one half of idiosyncratic passive variations.” Wealthier, better-educated, and better-diversified households adapt more aggressively toward their target allocations, while less-sophisticated households exhibit greater inertia. At higher frequencies, Gabaix et al. (2023) show that U.S. households’ trading responses to market movements are strongly wealth-dependent provided that less-affluent households tend to sell equities during downturns, whereas ultra-high-net-worth households buy and thus act counter-cyclically. Yet, despite these contrasting behaviors, the overall elasticity of household flows is small, limiting the aggregate stabilizing role of the household sector.

A complementary empirical literature highlights that wealthier households face systematically different return processes, which further shapes their adjustment dynamics. Using two decades of Norwegian administrative data, Fagereng et al. (2016) document substantial heterogeneity and persistence in returns to wealth, with a sizeable permanent component that “accounts for 60% of the explained variation.” They also show that returns rise with wealth, or “returns are positively correlated with wealth,” and that high-wealth households earn higher average and higher risk-adjusted returns, partly reflecting their greater exposure to private business assets and other high-return investments. These findings imply that both the scale of valuation shocks and the capacity to adapt portfolio shares differ across the wealth distribution, reinforcing the importance of studying group-level portfolio responses to unexpected changes in equity values.

These empirical patterns are central to modern heterogeneous-agent macroeconomics, where models of wealth inequality rely on persistent differences in returns, portfolio composition, and exposure to idiosyncratic capital-income risk. In the canonical framework of Benhabib and Bisin (2009), idiosyncratic and persistent heterogeneity in returns to wealth generates Pareto-tailed wealth distributions, with the right tail shaped primarily by capital-income risk rather than labor-income risk. Gabaix et al. (2015) further show that to explain the dynamics of top inequality, particularly the rapid rise observed in recent decades, models must incorporate size-dependent returns or heterogeneity in mean growth rates, mechanisms that closely mirror the empirical evidence on wealth-dependent returns and rebalancing behavior. Nirei and Aoki (2016) demonstrate that entrepreneurial productivity risk, combined with borrowing constraints and incomplete markets, naturally produces Pareto-tailed income and wealth distributions within a Bewley-type general equilibrium model. Their framework highlights how business-owner risk, precautionary savings, and tax policy jointly shape the upper tail, the channels emphasized in the micro evidence.

Taken together, this body of work shows that heterogeneity in returns, portfolio composition, and rebalancing behavior is not merely a micro-level curiosity but a central mechanism in macroeconomic theories of inequality. The empirical findings reviewed above therefore provide the micro-foundations needed to discipline heterogeneous-agent models and motivate examining how different segments of the wealth distribution adapt their portfolio shares after valuation shocks.

3 Data

The analysis uses the Distributional Financial Accounts (DFA) introduced by [Batty et al. \(2022\)](#), a quarterly dataset beginning in 1989Q3 that integrates the national-accounting structure of the Financial Accounts with the micro detail of the Survey of Consumer Finances (SCF). The DFA reports balance-sheet components for wealth groups of bottom 50%, 50–90%, 90–99%, 99–99.9%, and top 0.1%, and provides both levels and shares of major asset and liability categories reconciled to the Financial Accounts framework.

These data include group-level equity holdings, equity portfolio shares, fixed income, cash, private businesses, nonfinancial assets, and liabilities, which form the core variables in the empirical analysis. Because the DFA does not report market participation rates, changes in portfolio shares must be interpreted as group-level adjustments that combine both intensive-margin trading and extensive-margin entry or exit.

The project focuses on structural behavioral differences across groups, not on tracking the same individuals. Each wealth group represents a distinct economic environment, so individuals who move between groups are entering a different behavioral regime rather than contaminating the original one. Moreover, the local-projection framework estimates short-horizon responses (within 1 to 2 years), over which mobility across broad wealth quantiles is empirically limited. For these reasons, the DFA’s group-level aggregates remain the appropriate object.

The category “corporate equities and mutual fund shares” excludes private equity and private business assets. Because this category reflects public-market equity positions, the most coherent way to construct a passive benchmark is to apply a broad public-market return, even though wealth groups differ in how diversified their holdings are within this category. Methodological implications of this choice are discussed in detail in the subsequent section.

Before turning to the empirical strategy, the dataset will be explored descriptively. The DFA also provides distributions by education and age in addition to wealth percentiles. These alternative groupings offer additional context for understanding how portfolio composition and exposure to valuation shocks vary, possibly in light of financial literacy and life cycle aspects.

4 Methodology

4.1 Group-specific valuation shocks

Unexpected changes in equity values are identified using residuals from a predictive equation for the corresponding wealth accumulation process:

$$\Delta \ln L_{g,t} = \alpha_g + \phi_g \Delta \ln L_{g,t-1} (+\beta_g \bar{r}_t) + \varepsilon_{g,t},$$

where $L_{g,t}$ denotes the group’s equity holding levels and $\bar{r}_t$ is the overall market excess return adjusted by the risk-free rate. The residual $\varepsilon_{g,t}$ is interpreted as a group-specific valuation shock, capturing unexpected changes in equity values after accounting for market movements and past dynamics. This specification is intended to isolate “local” shocks that are specific to each group’s equity portfolio, net of aggregate market fluctuations.

To assess the sensitivity of the results to the shock definition, the analysis also considers a specification that omits the market-return term. Removing $\bar{r}_t$ yields a residual that reflects a broader equity-wealth shock, combining both market-driven and idiosyncratic components. This backup, simplest interpretation treats any unexpected change in equity wealth as the “global” disturbance.

4.2 Rebalancing Decomposition

An imperative piece of the literature is the distinction between passive drift i.e., mechanical changes in portfolio shares due to asset-price movements, and active rebalancing, which reflects deliberate trading behavior. To incorporate this distinction, a passive benchmark for the equity share is constructed by revaluing equity holdings using average market returns:

$$L_{g,t+h}^P = L_{g,t} \prod_{j=1}^h (1 + \bar{r}_{t+j}) .$$

The corresponding passive equity share is obtained by dividing this by the actual total assets observed in the DFA:

$$S_{g,t+h}^P = \frac{L_{g,t+h}^P}{A_{g,t+h}} .$$

This approach defines passivity relative to the realized evolution of the group's balance sheet, avoiding the pain to model group-specific total portfolio returns. Even though wealth groups differ in diversification and actual returns, the passive benchmark uses a common market return because it represents the mechanical effect of broad market movements. Subtracting this benchmark isolates the behavioral component of portfolio adaptation:

$$S_{g,t+h}^A = S_{g,t+h} - S_{g,t+h}^P .$$

This active rebalancing component measures how a group's equity share changes relative to the passive benchmark implied by market returns. A positive value indicates that the group increased its equity share more than passive drift would predict, while a negative value indicates active selling or a shift toward safer or more liquid assets. This interpretation holds regardless of whether average market returns are positive or negative: when returns are positive, the passive share rises mechanically, so a positive active component reflects reinforcing the boom; when returns are negative, the passive share falls mechanically, so a positive active component reflects cushioning the decline or buying into a downturn. In all cases, the sign of the active component cleanly captures the direction of deliberate portfolio adaptation relative to what would have happened had the group not traded at all as a counterfactual.

4.3 Core: Local Projections

Portfolio responses to valuation shocks are estimated using the basic local-projection framework of [Jordà and Taylor \(2024\)](#), applied separately to each wealth group. For each horizon h , the following specification is estimated:

$$S_{g,t+h} = \gamma_{g,h} + \delta_{g,h} \varepsilon_{g,t} + \rho'_{g,h} X_{g,t} + u_{g,t+h} ,$$

where $S_{g,t}$ is the equity portfolio share, $\varepsilon_{g,t}$ is the group-specific valuation shock, and $X_{g,t}$ includes its own lagged portfolio share ($S_{g,t-1}$) as well as balance-sheet controls in levels that mechanically affect portfolio shares in the short run (i.e., fixed income, private businesses, real estate, and liabilities).

Although some balance-sheet components such as pensions and miscellaneous assets are not modeled explicitly, their slow-moving nature and presence in the portfolio-share denominator mean that most of their influence is absorbed mechanically, while the included controls capture the major short-run, genuine rebalancing channels that could otherwise confound the estimated responses.

Estimating the specification separately for each group avoids imposing common dynamics and allows comparison of rebalancing speeds across the wealth distribution. The same specifi-

cation will be then estimated using the active rebalancing component $S_{g,t+h}^A$ as the dependent variable, isolating behavioral responses from passive valuation effects.

4.4 Asymmetric Shock Responses

To capture whether households react differently to gains and losses, the local-projection specification is augmented to allow separate coefficients for positive and negative valuation movements:

$$S_{g,t+h} = \gamma_{g,h} + \delta_{g,h}^+ \varepsilon_{g,t}^+ + \delta_{g,h}^- \varepsilon_{g,t}^- + \rho'_{g,h} X_{g,t} + u_{g,t+h},$$

where the shock components are defined as

$$\varepsilon_{g,t}^+ = \max(\varepsilon_{g,t}, 0), \quad \varepsilon_{g,t}^- = \min(\varepsilon_{g,t}, 0).$$

This formulation isolates whether negative shocks trigger stronger adjustments than positive shocks of the same magnitude, consistent with loss aversion. $|\delta_{g,h}^-| > |\delta_{g,h}^+|$ would indicate that groups reduce risk exposure more forcefully after losses, while similar coefficients would imply symmetric adjustment behavior.

4.5 Extended: Time-Varying Local Projections

Asset composition has evolved substantially over the sample period, alongside experiencing major financial disruptions. To capture potential shifts in rebalancing behavior, split-sample LPs are estimated:

$$S_{g,t+h} = \gamma_{g,h}^{(\tau)} + \delta_{g,h}^{(\tau)} \varepsilon_{g,t} + \rho_{g,h}^{(\tau)} X_{g,t} + u_{g,t+h}^{(\tau)},$$

where $\tau \in \{\text{pre-crisis, post-crisis}\}$ indexes the two subsamples before and after the 2008 financial crisis. This approach allows the impulse responses to differ flexibly without imposing parametric drift, providing insight into whether rebalancing speeds have changed as household balance-sheet structures evolved.

4.6 Alternative: Cash Share

To complement the analysis of equity rebalancing, an alternative set of local projections is estimated using the cash share as the outcome variable:

$$C_{g,t+h} = \eta_{g,h} + \theta_{g,h} \varepsilon_{g,t} + \varphi'_{g,h} Y_{g,t} + e_{g,t+h},$$

where $C_{g,t}$ denotes the cash share as a sum of money market fund shares and deposits. Cash represents households' primary liquid buffer, especially for lower-wealth groups, and may respond differently to valuation shocks.

4.7 Robustness: Panel LPs & BVARs

Two additional approaches are used to assess robustness. First, panel LPs introduce group fixed effects, which absorb time-invariant differences in portfolio levels and composition across wealth groups, and time fixed effects, which control for common macroeconomic conditions that affect all groups simultaneously. Second, Bayesian VARs provide a check on whether the joint dynamics of equity holdings and portfolio shares are consistent with the LP results under a richer dynamic structure with shrinkage accounting for heteroskedasticity.

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